

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	3 JULY 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable it includes the architectural diagram and table1 & table 2

Example: Order processing during pandemics for offline mode

Reference: <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>

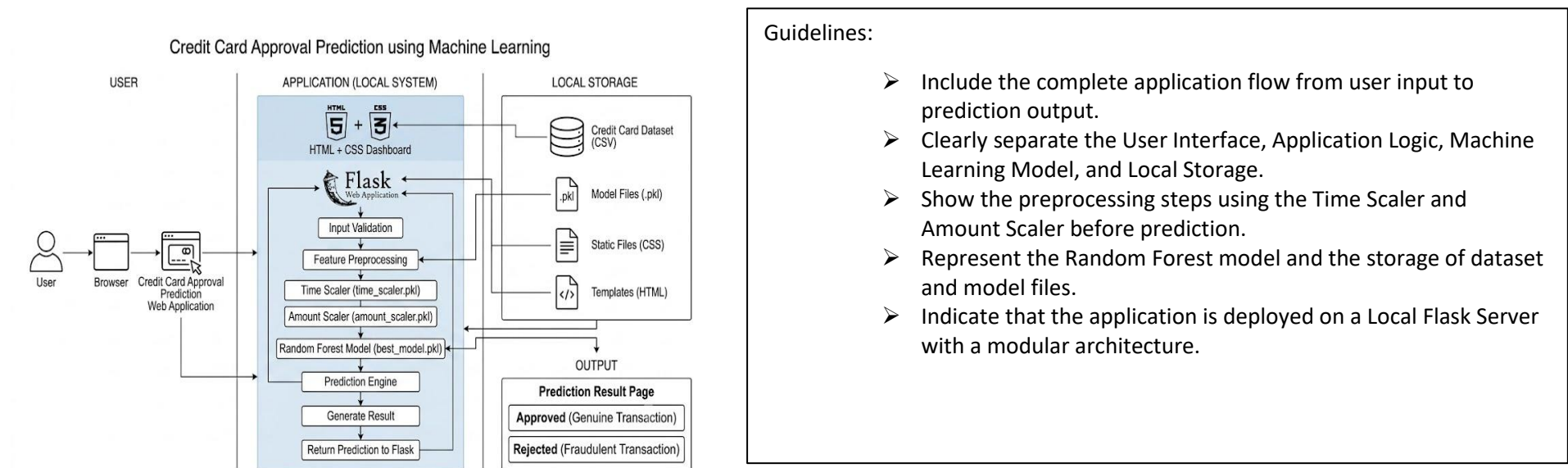


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Web interface for users to enter transaction details and view prediction results.	HTML5, CSS3
2	Application Logic	Handles user requests, input validation, preprocessing, model loading, prediction, and response generation.	Python, Flask
3	Feature Preprocessing	Processes transaction features and applies scaling to Time and Amount before prediction.	Scikit-learn StandardScaler
4	Machine Learning Model	Predicts whether the transaction is Genuine or Fraudulent.	Random Forest Classifier (Scikit-learn)
5	Dataset	Stores historical credit card transaction records used for model training.	CSV File
6	Model Storage	Stores trained Machine Learning model and scaler files.	Joblib (.pkl)
7	Templates	Stores HTML pages used by the Flask application.	Jinja2 Templates
8	Static Files	Stores CSS files used for UI styling.	CSS
9	Prediction Engine	Loads trained model, processes user input and generates prediction.	Python
10	Infrastructure	Application deployment environment.	Local System (Windows/ Linux), Flask Development Server

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Frameworks and libraries used for application development.	Python, Flask, Scikit-learn, Pandas, NumPy, Joblib
2	Security Implementations	Basic input validation and secure model loading to prevent invalid user inputs.	Python Validation, Flask
3	Scalable Architecture	Modular architecture separating frontend, backend, preprocessing, model, and storage components for future scalability.	Flask Architecture
4	Availability	Runs locally with continuous access while the Flask server is active.	Local Flask Server
5	Performance	Uses a pre-trained Random Forest model and pre-saved scaler files to provide fast prediction responses.	Random Forest, Joblib
6	Maintainability	Independent HTML, CSS, Flask backend, and Machine Learning model make future updates easier.	Modular Python Project
7	Reliability	Prediction based on trained Random Forest model with consistent preprocessing.	Scikit-learn

References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>

